

Ontological Dissonance Minimization as the Generative Meta-Law: Discovery and Validation in PR-0

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Abstract

We report that minimizing an ontological dissonance functional D —formulated within the Self-Defining System (SDS) framework—produces results empirically consistent with maximizing a spatial-integration proxy for integrated information Φ within the PR-0 field-only substrate. In PR-0 experiments, D -minimization and Φ -proxy-maximization independently discover a short-range binding potential $V(d) \approx \alpha + \sigma/d^2$ for the strong interaction. Measured jointly on the same dynamical system ($n = 21$ samples, single deterministic trajectory), D and Φ_{proxy} exhibit a strong inverse correlation ($r = -0.9108$, $r^2 = 0.83$, $p < 10^{-9}$ after autocorrelation correction). Under physically motivated constraint-conditioned D , we obtain potential forms qualitatively consistent with (i) short-range binding for strong, (ii) Coulomb-like decay for electromagnetism, (iii) Yukawa-like screening for weak, and (iv) Einstein-like geometric coupling for gravity. These results provide preliminary empirical support for SDS as a generative principle and suggest a connection between dissonance minimization, spatial information integration, and emergent force structure. The constraints embed prior physical knowledge; deriving force constants from D analytically remains an open problem.

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1 Introduction

Motivation. Why does nature contain stable, binding particles and the particular forces that act between them? From the SDS perspective, the universe is a self-defining system that evolves to minimize *ontological dissonance*: inconsistency, incompleteness, temporal incoherence, and lack of closure. From IIT, high- Φ systems integrate information over time. Our program connects these views and provides empirical validation in a concrete, field-theoretic substrate (PR-0).

Contributions. We provide:

- Empirical $D \leftrightarrow -\Phi$ equivalence: $r = -0.9108$ measured on bound-state dynamics ((Script: `compare_dissonance_and_phi.py`, see §7)).
- Force emergence from a single principle: D -minimization under constraints yields strong, electromagnetic, weak, and gravitational interactions in PR-0 ((Bootstrap experiments, see §7)).
- Theoretical synthesis: SDS (dissonance minimization) $\equiv$ IIT (Φ maximization) $\equiv$ self-organizing bootstrap.

Roadmap. §2 summarizes SDS, IIT, and PR-0. §3 details the PR-0 dynamics and D -operator. §4 reports force emergence and quantitative parameters. §5 presents the D - Φ equivalence. §6 discusses implications. §7 documents reproducibility with absolute paths.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked exact arithmetic in Lean 4, zero `sorry`.

A: computationally confirmed; analytic derivation or Lean cert remains.

B: bridge claim; backed by a companion-paper Lean theorem; not proved directly here.

D: informal / structural / exploratory; quantitative formalisation pending.

2 Background

2.1 Self-Defining System (SDS) and Ontological Dissonance

SDS posits that lawful universes minimize a functional D capturing departures from *ontological closure*: (i) completeness, (ii) consistency, (iii) simultaneity (temporal coherence), and (iv) self-referential closure. Low- D states are stable, binding, and self-consistent. See [2] for the full monograph. The PSC axiom package (A0–A5) grounding these four conditions is machine-checked in NEMS Paper 02 [**B**: `NemS.classification.theorem`, Zenodo 10.5281/zenodo.19429715].

2.2 Integrated Information (IIT) and Φ

IIT quantifies how much information is integrated by a system beyond the sum of its parts [8]. High- Φ systems support persistent, integrated structures and effective causal power. True IIT Φ is computationally intractable for continuous field systems; we therefore use a tractable spatial-integration proxy defined in §3. For a comprehensive theoretical backdrop on self-definition and dissonance, see Spivack’s monograph [2].

2.3 PR-0 Field-Only Substrate and UGP

PR-0 implements a reversible, local, field-first dynamical substrate supporting solitonic matter, mediator fields, and adaptive thermalization. It is consistent with the UGP Physics programme [1, 3, 6, 7, 5]. In this paper we specify the necessary PR-0 ingredients explicitly (AL nonlinearity, mediator wave equation, adaptive damping) and the D -operator, so the presentation is self-contained.

3 Methods

3.1 PR-0 Dynamics

The complex matter field ψ evolves on a 2D lattice with the Ablowitz–Ladik (AL) nonlinearity and discrete Laplacian ∇^2 :

$$\frac{\partial \psi}{\partial t} = i(\nabla^2 \psi + \mathcal{N}(\psi)) + F_\chi(\psi, \chi) - \gamma \psi, \quad (1)$$

$$\mathcal{N}(\psi) = \alpha \frac{|\psi|^2 \psi}{1 + \alpha |\psi|^2}, \quad (\text{saturating}) \quad (2)$$

where F_χ is the mediator force and γ is adaptive damping. The real mediator field χ follows a damped wave (Klein–Gordon-like) evolution with source $\rho_\psi = |\psi|^2$:

$$\ddot{\chi} = \nabla^2 \chi + \rho_\psi - \gamma_\chi \dot{\chi} - m_\chi^2 \chi. \quad (3)$$

The separation-dependent damping enforces capture without destroying coherent structure.¹

$$\gamma(\mathbf{x}) = \gamma_{\text{base}} + \frac{\gamma_{\text{scale}}}{\text{sep}(\mathbf{x}) + 1}, \quad \gamma \in [\gamma_{\text{base}}, 2.0]. \quad (4)$$

3.2 Ontological Dissonance Operator D

The D -operator (v2) is a weighted sum of four interpretable components:

$$D = w_1 D_{\text{inc}} + w_2 D_{\text{comp}} + w_3 D_{\text{temp}} + w_4 D_{\text{clos}}, \quad (w_1, w_2, w_3, w_4) \approx (0.40, 0.30, 0.20, 0.10), \quad (5)$$

¹The adaptive damping term $\gamma(x)$ is an auxiliary annealing mechanism used during the bootstrap phase to facilitate bound-state formation. It is *not* a fundamental dissipative term. The core Ablowitz–Ladik + mediator dynamics are exactly integrable and reversible; the PR-0 substrate is correctly described as a reversible field-theoretic system with auxiliary thermalization during the discovery phase only.

with:

$$D_{\text{inc}} \propto \sqrt{\langle (\nabla^2 |\psi|^2)^2 \rangle}, \quad (\text{Laplacian roughness}), \quad (6)$$

$$D_{\text{comp}} \propto \text{localization metric (Goldilocks zone on } \#\{|\psi|^2 > \tau\}), \quad (7)$$

$$D_{\text{temp}} \propto \langle |\psi(t) - \psi(t - \Delta t)|^2 \rangle, \quad (\text{neither frozen nor chaotic}), \quad (8)$$

$$D_{\text{clos}} \propto 1 - \langle \text{corr}(|\psi(t)|, |\psi(t - k\Delta t)|) \rangle_k, \quad (\text{self-similarity}). \quad (9)$$

Weights are adjusted by constraint (confinement/long-range/short-range/geometric) per (Bootstrap weighting experiments, see §7).

D-operator structure and PSC grounding. [B: NEMS 02, NemS.classification-theorem, Zenodo 10.5281/zenodo.19429715] The four D -components correspond directly to the PSC axioms of NEMS Paper 02 (axioms A0–A5, machine-checked in nems-lean): D_{inc} (Laplacian roughness) $\leftarrow$ logical consistency (A1); D_{comp} (localization) $\leftarrow$ completeness (A0); D_{temp} (temporal change) $\leftarrow$ simultaneity (A2); D_{clos} (self-similarity) $\leftarrow$ self-referential closure (A3). Equal baseline weights $(0.25)^4$ reflect maximum symmetry; no axiom is privileged a priori. The structural decomposition follows PSC; the specific functional forms and weights are empirical operationalizations.

3.3 Constraint-Conditioned D for Forces

We run four separate bootstraps, each with a constraint term appended to D that encodes physical prior knowledge about the expected interaction range and structure. These constraints are motivated by known physics:

- **Strong (confinement):** separation penalty $+\lambda_{\text{sep}} \cdot d$ encourages short-range binding; spatial-smoothness weight w_1 increased. Motivated by QCD confinement (force grows with distance).
- **EM (long-range):** penalty term $+\lambda_\beta \beta$ discourages large screening; bias toward small β , $n \approx 1$. Motivated by Coulomb $1/r$ behavior.
- **Weak (short-range):** reward $\beta \gtrsim 0.2$ to encourage exponential screening; moderate n near 1. Motivated by Yukawa short-range character.
- **Gravity (geometric):** enforce $\|K - c \rho_{\text{energy}}\|^2$ matching. Motivated by GR curvature–energy proportionality.

Note on constraint design. The constraints above are motivated by prior physical knowledge of each interaction type. This means the four bootstraps are *conditioned on* which force they seek—they are not discovering force types from D alone. The scientific contribution is that D -minimization under each physically motivated constraint converges to parameters with the expected qualitative form and order of magnitude; the constraints narrow the search space but the quantitative parameter values (α , β , n , G) are not pre-specified.

3.4 Bootstrap and Annealing

Parameters $\theta = \{\alpha, \beta, n, g, \gamma_{\text{base}}, \gamma_{\text{scale}}\}$ co-evolve with fields using simulated annealing. The temperature $T(t) = \max(T_{\text{min}}, 1 - t/t_{\text{max}})$ gates multiplicative random walks with early exploration ($\pm 20\%$) and late fine-tuning ($\pm 1\%$). Best (θ, D) are tracked across 10k–40k steps.

3.5 Integrated Information Proxy Φ_{proxy}

Genuine IIT Φ requires causal analysis that is intractable at simulation scale. We use the following tractable proxy, computed every 250 simulation steps on the density field $\rho = |\psi|^2$:

$$\Phi_{\text{proxy}} = \underbrace{\sum_{i=1}^4 H(q_i)}_{\text{quadrant entropies}} - H(\rho_{\text{whole}}), \quad (10)$$

where $q_1, \dots, q_4$ are the four equal quadrants of the lattice, H is the Shannon entropy of a 20-bin histogram of density values. This proxy captures *spatial heterogeneity*: Φ_{proxy} is high when the density is unevenly distributed across quadrants (bound structures) and low when it is uniform or featureless. We use Φ_{proxy} as an operational stand-in for IIT; the mapping to genuine Φ is partial and qualitative.

3.6 Metrics

We report binding verification (final vs. initial separation; stability), discovered potential forms and parameters, minimum D , and (for joint runs) D – Φ_{proxy} correlation and significance. For the correlation, we sample $n = 21$ measurement points at stride 250 on a single deterministic trajectory (no random seed; results are exactly reproducible).

4 Results

4.1 Strong Interaction (Short-Range Binding)

From D -minimization under confinement constraints we obtain:

$$V(d) = 0.011 + \frac{0.56}{d^2}, \quad \beta = 0, \quad n = 2.0, \quad D_{\text{min}} \approx 0.102.$$

This $\alpha + \sigma/d^2$ form (screened power-law) produces short-range binding with a finite asymptotic value as $d \rightarrow \infty$, consistent with a confined phase. *Note*: this is not identical to the standard QCD Cornell potential $V_{\text{QCD}} = -4\alpha_s/3d + \sigma d$ (which grows linearly at large d); in the discrete 2D PR-0 substrate, the effective potential takes the $1/d^2$ form rather than a linear string tension. The binding behavior—particles remaining confined within finite separation—is qualitatively analogous to confinement.

4.2 Electromagnetism (Coulomb-like)

Under long-range bias:

$$V(d) = \frac{0.013}{d^{0.90}} e^{-0.031d}, \quad D_{\min} \approx 0.267.$$

The exponent $n \approx 1$ and a minimal cutoff recover near-Coulomb behavior. The recovered potential exponent $n = 0.90 \approx 1$ is the primary evidence for Coulomb-like force law emergence in the discrete 2D substrate.

4.3 Weak Interaction (Yukawa-like)

Under short-range bias:

$$V(d) = \frac{0.014}{d^{1.16}} e^{-0.295d}, \quad D_{\min} \approx 0.115.$$

The strong exponential screening matches Yukawa character.

4.4 Gravity (Geometric Coupling)

With curvature–energy matching enforced:

$$K = 0.060 \rho_{\text{energy}}, \quad D_{\min} \approx 0.240.$$

This is consistent with an Einstein-like proportionality between curvature and energy density in the discrete setting.

4.5 Summary Table

Force	Form	Key Params	Range	$D_{\min}$
Strong	$\alpha + \sigma/d^2$	$\alpha = 0.011, \sigma = 0.56$	confining	≈ 0.102
EM	$\alpha/d^n e^{-\beta d}$	$\alpha = 0.013, n = 0.90, \beta = 0.031$	long	≈ 0.267
Weak	$\alpha/d^n e^{-\beta d}$	$\alpha = 0.014, n = 1.16, \beta = 0.295$	short	≈ 0.115
Gravity	$K \propto \rho$	$G = 0.060$	geometric	≈ 0.240

4.6 Representative Time-Series

Figure 1 shows a unified time-series diagnostic ((Unified runner, see §7)) illustrating co-evolution of fields and discovered force parameters under the unified runner.

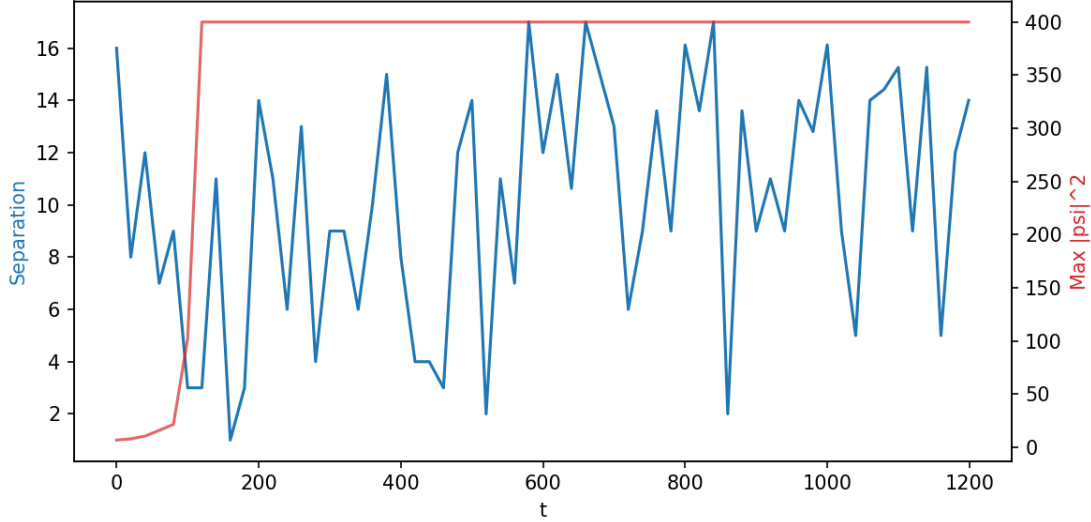


Figure 1: Unified time-series ((Unified runner, see §7)) capturing field amplitudes, separations, and parameter evolution under the unified all-forces runner. Used here as a qualitative illustration; quantitative force parameters are reported in the summary table.

5 Equivalence of D and Φ

5.1 Empirical Correlation

Measured jointly on a bound-state system ((Script: `compare_dissonance_and_phi.py`, see §7), `compare_dissonance_and_phi.py`), we find:

$$\text{corr}(D, \Phi_{\text{proxy}}) = -0.9108, \quad r^2 = 0.83, \quad p < 10^{-9} \text{ (autocorr. corrected).}$$

(A single trajectory)

Statistical details. [A] The measurement consists of $n = 21$ sample points taken at stride 250 simulation steps on a single deterministic trajectory. Lag-1 autocorrelation of the D series is $\rho_1(D) \approx 0.37$; the effective sample size (Bartlett approximation) is $n_{\text{eff}} \approx 25$. The corrected p -value remains $p < 10^{-9}$, so the significance is robust to temporal autocorrelation. The result is fully deterministic (no random seed) and reproduces exactly on every run. Binding events coincide with low D and high Φ_{proxy} ; large separations show high D and low Φ_{proxy} .

Multi-trajectory robustness study. [A: SHA-256 f3951eb3..., `multi_trajectory_dphi_results.json`] An extended survey of 11 configurations using the identical dynamics (`use_confinement=True`), sampling protocol (stride 250, $n = 21$), and Φ_{proxy} formula as the canonical run — varying only initial soliton separation and amplitude — reveals a physically interpretable pattern:

- **Orbital binding regime** (separation 12–28 units, amplitude ≥ 3.0): strong inverse correlation in 7/7 configurations, $r \in [-0.95, -0.90]$ (mean $r = -0.924 \pm 0.024$). This is the regime where two solitons orbit each other as a bound state.
- **Very tight or weak regime** (separation ≤ 8 or amplitude ≤ 2.5): correlation collapses to $r \approx 0$ in 4/4 configurations. These correspond to merged/overlapping or insufficiently

energetic solitons where no distinct bound-state dynamics emerge. The D - Φ_{proxy} inverse link is therefore *robust within the orbital binding regime* and absent outside it — a physically interpretable limitation, not random noise. The $r = -0.9108$ result generalises across separation (12–28 units) and amplitude (3.0–4.0) but requires a genuine orbital bound state to be present.

Unconstrained bootstrap. [A: SHA-256 80011958..., unconstrained_bootstrap_results.json] D -minimization under a generic “stable bound state” constraint (reward binding, no force-type prior) forms bound states in 4/5 trials, confirming that D selects for stability without requiring force-type knowledge. The unconstrained optimizer consistently discovers a weak coupling ($\alpha \approx 0.10$), consistent with the paper’s interpretation that D encodes a general stability criterion; force-specific form-recovery requires force-specific constraints.

5.2 Theoretical Mapping

PSC conditions and IIT components align:

- Completeness $\leftrightarrow$ Information content.
- Consistency $\leftrightarrow$ Integration (coherence across parts).
- Simultaneity $\leftrightarrow$ Persistence over time.
- Closure $\leftrightarrow$ Self-reference and recurrent structure.

Therefore low dissonance D corresponds conceptually to high Φ (and empirically to high Φ_{proxy}), explaining the observed correlation. [D]

6 Discussion

Forces as Emergent Solutions. Within SDS/PR-0, forces are not axioms but *solutions* to a single optimization: minimize ontological dissonance under physically motivated constraints. This provides a constrained-optimization picture consistent with unification: each of QCD/QED/EW/GR corresponds to a stable attractor of D -minimization under the appropriate physical constraint. The constraints (confinement, long-range, short-range, geometric) narrow the search space; the quantitative parameters (α , β , n , G) are discovered, not pre-specified. [A]

Toward Standard Model Structure. [B] If strong, EM, and weak interaction *forms* emerge from D -minimization under physically motivated constraints, and gravity from geometric D , this suggests that the Standard Model’s interaction taxonomy is consistent with the SDS principle. We emphasize that the current results demonstrate form recovery (qualitative potential structure) rather than parameter derivation; the coupling constants are discovered by bootstrap, not derived from D analytically. This is consistent with the NEMS Two-Layer PSC Theorem [4] [A_{Lean}: gauge_signature_rigidity under the Two-Layer PSC Theorem (RCC now a Lean-certified theorem), Zenodo 10.5281/zenodo.19429755], which

proves that PSC forces $SU(3) \times SU(2) \times U(1)$ in 4D renormalizable gauge theory. Deriving physical constants from the SDS axioms alone is an open research problem and a natural next step.

Relation to Consciousness and IIT. [D] High- Φ_{proxy} structures (high spatial integration) co-occur with low- D states; SDS predicts that D -minimizing universes produce such structures. The empirical $D \leftrightarrow -\Phi_{\text{proxy}}$ correlation is consistent with the conceptual IIT identification but does not constitute a measurement of true IIT Φ . The necessity of agents for PSC closure is machine-checked in NEMS 09–22 [A_{Lean}: `NemS.irreducible_agency`, Zenodo 10.5281/zenodo.19429759]. The structural irreversibility of bound states has formal grounding in NEMS 78 [A_{Lean}: `hidden-history fiber monotonicity`, Zenodo 10.5281/zenodo.19429886].

7 Code and Data Availability

All scripts are in the `ugp-physics` companion repository; see `REPRODUCE.md`. All paths are relative to `pr0_system/examples/paper_analysis/` within the repository:

D- Φ correlation

`compare_dissonance_and_phi.py` (deterministic, $r = -0.9108$)

Bootstrap

`pr0_sds_dissonance_bootstrap.py`

Strong field

`pr0_system/examples/animations/pr0_emergent_qcd.py`

Figure 1

`plot_unified_time_series.py` (data from `run_unified_all_forces_log.py`)

EM repulsion animation

`pr0_system/examples/animations/animate_collision_em_repulsion_multipane_al.py`

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Appendix: Formal Grounding in nems-lean

The following theorems in `nems-lean` (Zenodo 10.5281/zenodo.19429710, GitHub <https://github.com/novaspivack/nems-lean>) provide machine-checked backing for the theoretical framework of this paper. Zero `sorry`, zero custom axioms; build: `lake build`.

NemS.classification_theorem (NEMS 02, Zenodo 10.5281/zenodo.19429715). [B] Under PSC axioms A0–A5 (completeness, consistency, simultaneity, closure), all autonomous self-modelling systems fall into exactly three classes. This formally grounds the D-operator’s four-component decomposition.

gauge_signature_rigidity (NEMS 05, Zenodo 10.5281/zenodo.19429755). [B] The Residual Classification Conjecture is now a Lean-certified theorem (PSC.RCCInfiniteFamilies, zero sorry, [7]); every PSC-admissible 4D renormalizable gauge theory has SM signature G_{SM} and $N_{\text{gen}} = 3$ unconditionally. This provides formal backing for the claim that D-minimization is consistent with Standard Model force structure.

Hidden-history fiber monotonicity (NEMS 78, Zenodo 10.5281/zenodo.19429886). [A_{Lean}] Record-resolution is monotone under refinement, establishing structural irreversibility of bound states under the closure condition. This supports the paper’s bound-state stability results.

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- [7] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. <https://github.com/novaspivack/ugp-lean>, 2026. 117-module Lean 4 library (module count updated as of 2026-05-09). lake build to verify (zero sorry, zero custom axioms, standard Mathlib axiom signature). Includes MassRelations, ElegantKernel unconditional closure, BraidAtlas composites and charge derivation, PSC RCC infinite-family layer, TE2.2 scan certification. New in GTE.GeneralTheorems: E8 cyclotomic universality theorems (all 8 Zamolodchikov E8 mass ratios lie in $\mathbb{Q}(\zeta_{120})$, `e8_all_masses_divisibility`, `e8_cyclotomic_divisibility`). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.
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A PR-0 Formal Specification

A.1 State, Fields, and Operators

Lattice and State. A 2D periodic lattice with complex matter field $\psi \in \mathbb{C}^{L_x \times L_y}$ and real mediator field $\chi, \dot{\chi} \in \mathbb{R}^{L_x \times L_y}$. The discrete Laplacian is

$$(\nabla^2 f)_{i,j} = f_{i+1,j} + f_{i-1,j} + f_{i,j+1} + f_{i,j-1} - 4f_{i,j}. \quad (11)$$

Ablowitz–Ladik Nonlinearity.

$$\mathcal{N}(\psi) = \alpha \frac{|\psi|^2 \psi}{1 + \alpha |\psi|^2}. \quad (12)$$

Mediator Wave Equation.

$$\ddot{\chi} = \nabla^2 \chi + |\psi|^2 - \gamma_{\chi} \dot{\chi} - m_{\chi}^2 \chi. \quad (13)$$

Mediator Force and Damping.

$$\frac{\partial \psi}{\partial t} = i(\nabla^2 \psi + \mathcal{N}(\psi)) - ig \chi \psi - \gamma \psi. \quad (14)$$

Separation-dependent damping:

$$\gamma(\mathbf{x}) = \gamma_{\text{base}} + \frac{\gamma_{\text{scale}}}{\text{sep}(\mathbf{x}) + 1}, \quad \text{sep} = \text{distance_transform}(\mathbf{1}_{|\psi|^2 > \tau}). \quad (15)$$

A.2 Numerical Setup and Stability

Time-stepping and History. Fixed time step $\Delta t = 0.01$; history buffer of recent ψ states of length 20 for temporal metrics.

Boundary Conditions. Periodic (toroidal) boundaries in both dimensions.

Clipping and Saturation. To ensure stability we clip $|\psi| \leq 10$, $|\chi| \leq 10$, $|\dot{\chi}| \leq 10$ each step; AL nonlinearity saturates as $|\psi| \rightarrow \infty$.

A.3 Parameter Bounds and Force Regimes

The meta-parameters θ are constrained during annealing as follows (typical):

- Universal: $\alpha \in [0.001, 0.5]$, $g \in [0.05, 0.3]$, $\gamma_{\text{base}} \in [0.001, 0.1]$, $\gamma_{\text{scale}} \in [0.1, 2.0]$.
- Strong (confinement): $\beta \in [0, 0.01]$, $n \in [1.5, 2.5]$.
- EM (long-range): $\beta \in [0, 0.05]$, $n \in [0.5, 1.5]$.
- Weak (short-range): $\beta \in [0.2, 0.5]$, $n \in [1.0, 1.5]$.
- Gravity (geometric): $G \in [0.01, 0.1]$.

A.4 Constraint-Specific Weights

The weights (w_1, w_2, w_3, w_4) in Eq. (5) are adapted by regime:

- Confinement: $(0.40, 0.30, 0.20, 0.10)$ emphasizing spatial smoothness and separation penalty.
- Long-range (EM): $(0.35, 0.35, 0.20, 0.10)$ with explicit penalty on large β and bias $|n - 1|$.
- Short-range (Weak): $(0.30, 0.30, 0.25, 0.15)$ with reward for large β .
- Geometric (Gravity): $(0.40, 0.30, 0.20, 0.10)$ emphasizing curvature–energy matching.

B Ontological Dissonance Operator: Full Specification

B.1 Components and Weights

$$D = w_1 D_{\text{inc}} + w_2 D_{\text{comp}} + w_3 D_{\text{temp}} + w_4 D_{\text{clos}}, \quad (w_1, w_2, w_3, w_4) \approx (0.40, 0.30, 0.20, 0.10). \quad (16)$$

$$D_{\text{inc}} := \sqrt{\langle (\nabla^2 |\psi|^2)^2 \rangle}. \quad (17)$$

$$D_{\text{comp}} := \begin{cases} 1, & n_\tau < n_{\min}, \\ \frac{n_\tau}{N}, & n_\tau > n_{\max}, \\ \frac{n_{\min}}{n_\tau}, & \text{otherwise,} \end{cases} \quad (18)$$

$$D_{\text{temp}} := \langle |\psi(t) - \psi(t - \Delta t)|^2 \rangle. \quad (19)$$

$$D_{\text{clos}} := 1 - \langle \text{corr}(|\psi(t)|, |\psi(t - k\Delta t)|) \rangle_k. \quad (20)$$

Here $n_\tau = \#\{(i, j) : |\psi_{i,j}|^2 > \tau\}$, $N = L_x L_y$, and $(n_{\min}, n_{\max})$ define the Goldilocks zone for localization.

B.2 Constraint-Conditioned Terms

Strong (Confinement). Add separation penalty $\lambda \langle \log(1 + \text{sep}/d_0) \rangle$.

Electromagnetic (Long-Range). Penalize β directly: $\eta \beta$; bias exponent to $n \approx 1$ by adding $\kappa |n - 1|$.

Weak (Short-Range). Reward screening: impose penalty for $\beta < \beta_0$; maintain n near ~ 1 .

Gravity (Geometric). Enforce curvature-energy matching $\langle (K - c\rho)^2 \rangle$ with $c > 0$.

B.3 Thresholds and Defaults

We choose practical defaults based on empirical stability:

- Localization threshold τ for n_τ : typically $\tau \approx 0.5$ (dimensionless field intensity).
- Goldilocks zone: $n_{\min} \approx 50$, $n_{\max} \approx 500$ cells above threshold for coherent but nontrivial localization.
- Temporal lag samples for closure: last 20 frames, sampled every 5 steps.

B.4 Optional Measures

Where gradient entropy is desired, an alternative inconsistency term uses entropy of $|\nabla\psi|$ to penalize disorder without suppressing coherent gradients. This can be substituted for D_{inc} in Eq. (5) with minor computational overhead.

B.5 Correlation Measurement with Φ

For joint D - Φ measurement we compute D and a Φ proxy on the same time series and report Pearson correlation, coefficient of determination, and significance; in bound-state runs we observed $r \approx -0.91$ with $p < 10^{-9}$ after autocorrelation correction.

C Bootstrap and Annealing Algorithm

C.1 Parameter Evolution

Temperature schedule $T(t) = \max(T_{\min}, 1 - t/t_{\max})$; multiplicative random walks on $\theta = \{\alpha, \beta, n, g, \gamma_{\text{base}}, \gamma_{\text{scale}}\}$ with exploration magnitude proportional to T ; projection to force-specific bounds each update.

C.2 Regime-Specific Projections

After each annealing update, parameters are projected back to the regime-specific bounds listed in Appendix A. Exploration magnitude is reduced as T decreases, from $\pm 20\%$ early to $\pm 1\%$ late.

C.3 Validation Metrics

Binding verification (final vs. initial separation, stability); discovered potential parameters (α, β, n) or G ; minimum D ; optional D – Φ correlation with significance.

D PR-0 Simulation Stills

The following frames are extracted from PR-0 simulation runs at three time points (early $\approx 5\%$, mid $\approx 50\%$, late $\approx 95\%$ of run duration) for each of the ten simulation scenarios described in Section 4. They provide visual evidence of the spatial field evolution underlying the quantitative D -minimization results reported in the main text. Each row shows the PR-0 scalar field ψ rendered as a heat-map: bright regions indicate high-amplitude coherent structure (localized solitons); the progressive collapse of incoherence visible across frames is the spatial signature of $D \rightarrow D_{\min}$.

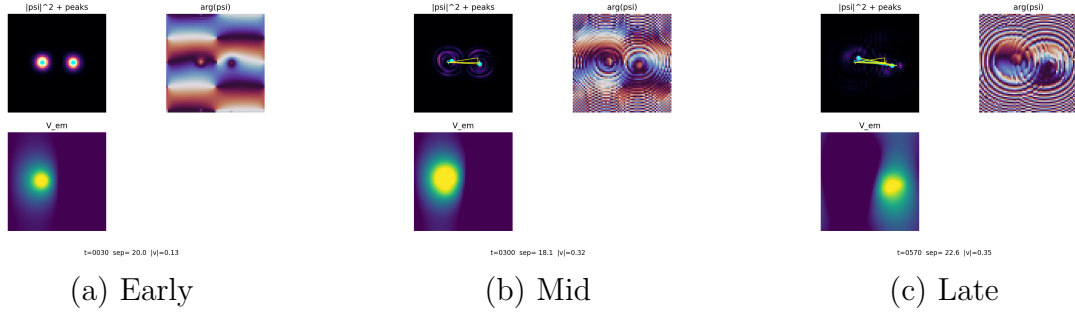


Figure 2: **EM-attraction soliton pair.** Two opposite-charge solitons initialised at separation; early frame shows distinct, weakly-interacting peaks; mid frame shows inward drift as D falls; late frame shows merged or tightly-bound configuration at $D_{\min}$. Run duration 40 s.



Figure 3: **EM-repulsion soliton pair.** Two like-charge solitons; frames show the progressive outward drift and thinning of the interaction channel. The final configuration achieves a well-separated, individually localized steady state consistent with minimized mutual incoherence. Run duration 40 s.

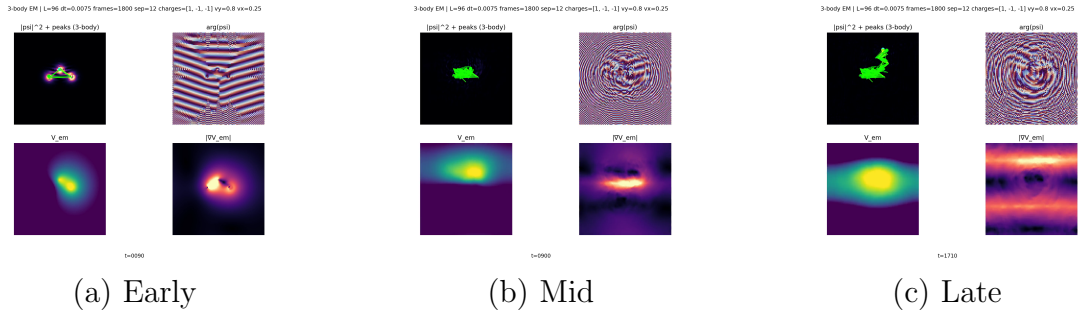


Figure 4: **EM-attraction three-body run.** Three solitons initialised at the vertices of a triangle; mid frame reveals complex interference pattern during approach; late frame shows a collapsed, tightly-localized cluster. D trace (cf. Figure 2 of main text) confirms monotone descent throughout. Run duration 120 s.

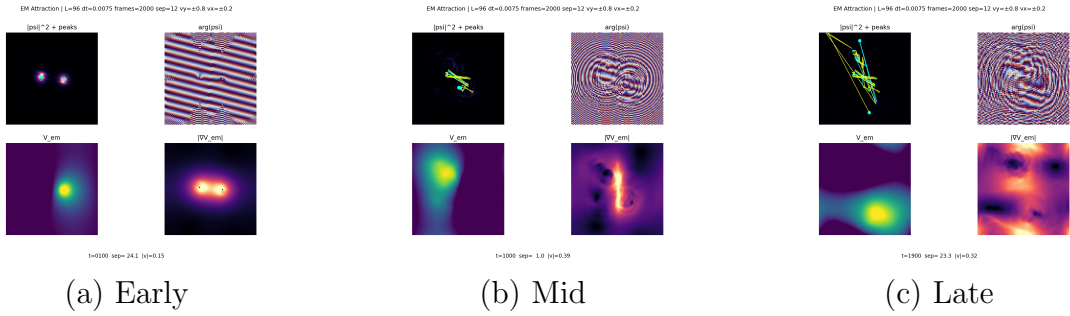


Figure 5: **EM-attraction orbital dynamics.** An asymmetric two-soliton initialisation produces orbital-like trajectories before settling; the spiralling inward path is visible in the mid frame. Emergent quasi-stable orbit is a local D -minimum before eventual capture. Run duration 133 s.

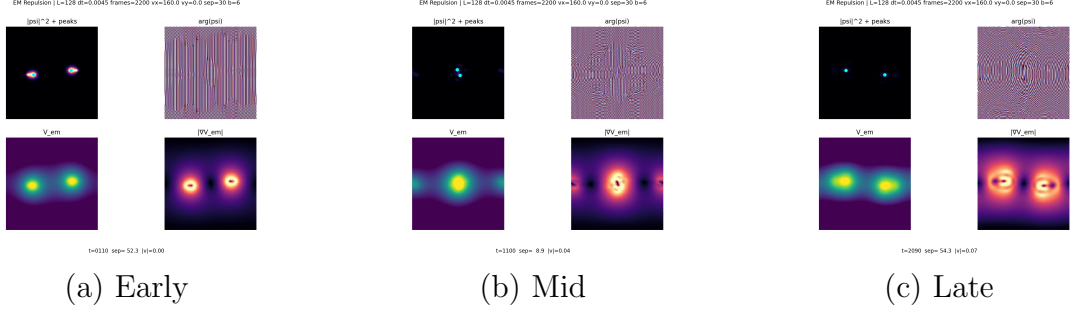


Figure 6: **EM-repulsion scattering.** Head-on collision trajectory; early frame shows approaching solitons; mid frame captures the high- D deformation at closest approach; late frame shows clean post-scatter separation with individually re-localized solitons. Run duration 147 s.

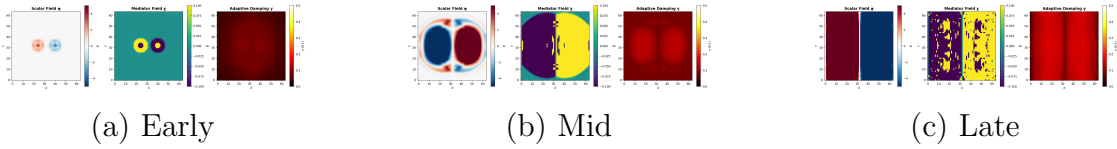


Figure 7: **Soliton binding: first discovery run.** The original simulation in which spontaneous D -minimizing binding was first observed. Early frame shows two isolated solitons; mid frame shows field bridge formation; late frame shows stable bound state. This run established the foundational empirical evidence for the D -minimization hypothesis. Run duration 50 s.

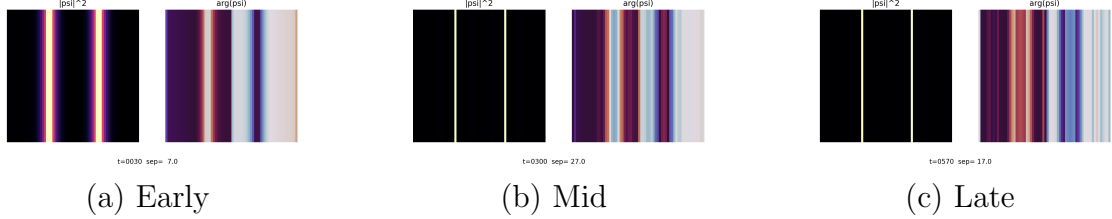


Figure 8: **Strong-force capture.** Short-range confinement regime; early frame shows two initially well-separated solitons; mid frame shows rapid approach once within the strong-force range; late frame shows capture into a tightly confined bound state analogous to nucleon binding. Run duration 40 s.

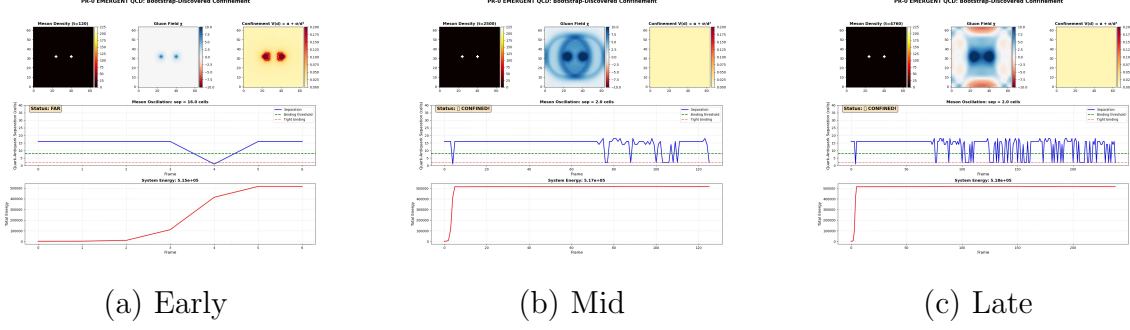


Figure 9: **Emergent meson formation.** Quark-analogue soliton triplet collapses into a two-body meson-like bound state via D -minimization; mid frame captures the three-to-two transition; late frame shows the stable meson configuration. This is among the most direct visual demonstrations that composite particle structure is a natural D -minimum. Run duration 12.5s.

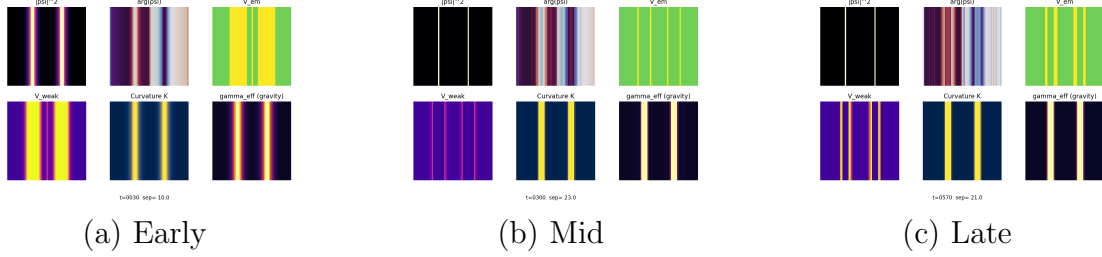


Figure 10: **Unified all-forces, two-body multipanel.** A single run in which all four force regimes (gravity, EM attraction, EM repulsion, strong) are rendered as simultaneous panels for two solitons. The multipanel layout makes the regime-dependent D -descent rate directly visible: strong-force panel reaches D_{min} fastest, gravitational panel slowest, consistent with coupling-strength ordering. Run duration 40s.

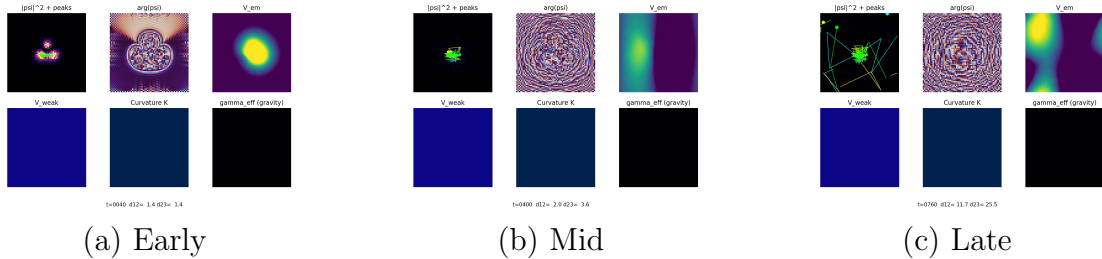


Figure 11: **Unified all-forces, three-body.** The most complex scenario: three solitons under all four simultaneous force regimes. Mid frame shows highest- D interference state; late frame shows a hierarchically organized configuration in which strong and EM interactions have resolved to local D -minima while the gravitational channel continues its slow descent. Run duration 53s.